

Compressed Tries

1 Definition

Compressed Trie

A **compressed trie** removes unnecessary chains of one-child internal nodes. Instead of storing exactly one character per transition, it stores a substring on an edge.

The defining structural rule is:

Every internal node has at least two children.

This means nonbranching paths are merged into single substring-labelled edges.

2 Redundant Internal Nodes

Redundant Node

An internal node v is redundant when:

- v has exactly one child;
- v is not the root.

Such a node does not represent a branching decision. It can be removed by combining the incoming and outgoing edge labels.

3 Chain Compression

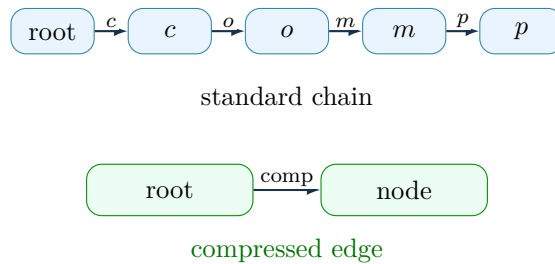
Consider a chain

$$(v_0, v_1), (v_1, v_2), \dots, (v_{k-1}, v_k).$$

The intermediate nodes $v_1, \dots, v_{k-1}$ are redundant if each has exactly one child. The whole chain is replaced by one edge:

$$v_0 \longrightarrow v_k.$$

The new edge stores the combined substring.



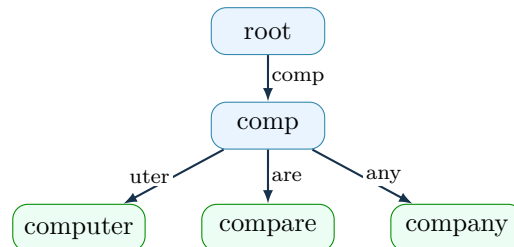
4 Common Prefix Example

For

computer, compare, company,

all three words begin with

comp.



5 Larger Example

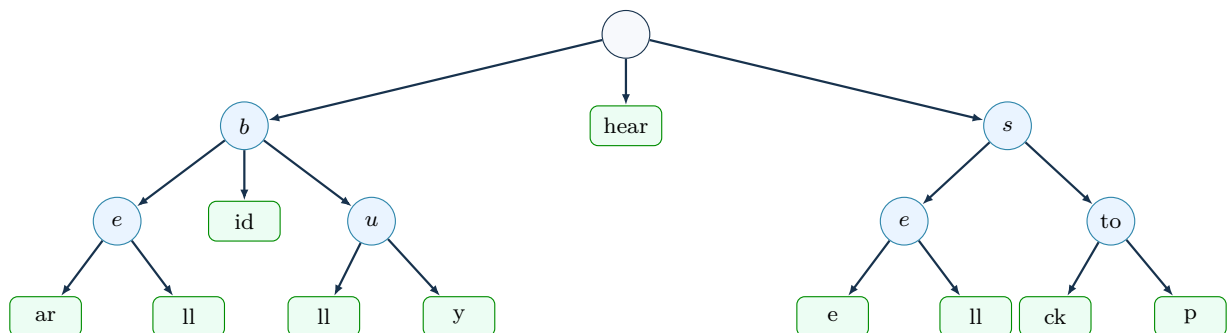
For

{see, bear, sell, stock, bull, buy, hear, bell, bid, stop},

the compressed trie can be read by top-level branches:

root branches: <i>b</i> , <i>h</i> , <i>s</i> .

Branch	Compressed continuation
<i>b</i>	$e \rightarrow ar$ for bear, $e \rightarrow ll$ for bell, id for bid, $u \rightarrow ll/y$ for bull/buy
<i>h</i>	<i>hear</i> is stored as one compressed edge
<i>s</i>	$e \rightarrow e/ll$ for see/sell, and $to \rightarrow ck/p$ for stock/stop



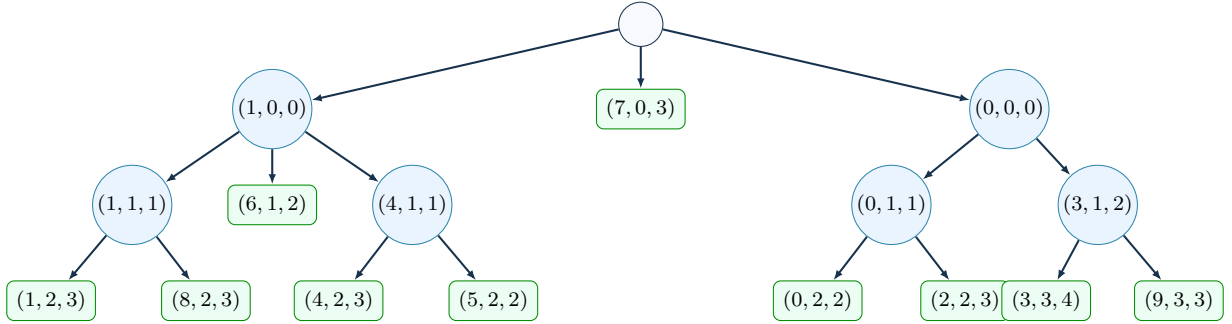
Reading the diagram

Labels such as *id*, *ar*, and *to* are compressed edge labels. They replace chains of one-character nodes.

6 Compact Index Representation

Instead of storing edge labels as strings, store references into the original string array.

Index	String	Index	String
$S[0]$	see	$S[5]$	buy
$S[1]$	bear	$S[6]$	bid
$S[2]$	sell	$S[7]$	hear
$S[3]$	stock	$S[8]$	bell
$S[4]$	bull	$S[9]$	stop



Meaning of (i, j, k)

The label (i, j, k) means the edge stores substring $S[i][j..k]$. Example: if $S[1] = \text{bear}$, then $(1, 2, 3)$ represents ar.

7 Search Behavior

When searching in a compressed trie, compare the query against whole edge labels:

edge label = substring, not one character.

Do Not Skip Edge Checks

If an edge is labelled comp, the query must match all four characters c, o, m, p before moving to the next node.